Yuyuan Han

Computational imaging & robust image-free sensing

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PhD researcher in single-pixel imaging and image-free sensing, working toward robot AI that reads the physics in raw sensor data: designs optical measurements and trains networks that act directly on them.

Education & Research Appointments

Xi'an Jiaotong University · PhD student — 2022–present
Electronic Science and Technology · Master's-to-PhD track · Advisor: Huaibin Zheng

Politecnico di Milano · Visiting PhD student — Oct 2026–present
Department of Physics · Supervisor: Antonio Pifferi

Northwest University · B.Eng. — 2018–2022
Electronic Information Science and Technology

Selected Research

The Lift Spectrum · Robust image-free sensing — Preprint

arXiv · 2026 · Paper · Code

- Showed that without reconstruction the 1D-to-2D lift is the design problem; mapped the lift spectrum.
- Proposed STSF + TPLS: +3.2–9.9 pp over the prior image-free method (SPIFS); near-constant accuracy down to 0.78% sampling; up to +21.7 pp over clean-trained reconstruct-then-segment at 20 dB noise.

Sparse Hadamard single-pixel imaging

Optics Express · 2025 · Paper · Code

- Gave orthogonal Hadamard patterns tunable sparsity: noise-robust imaging through scattering, low-order matrices that save memory, no training or post-processing.

LDGI · Optical phased-array imaging

Optics Communications · 2025 · Paper · Code

- Showed that line patterns from one-dimensional optical phased arrays break the noise symmetry of ghost-imaging denoising; proposed LDGI (orders-of-magnitude gains), validated optically.

Current Research

Acquisition-aware single-pixel sensing with optical-bench tests (manuscript in preparation); language tasks from raw single-pixel measurements (exploratory). Planned at Politecnico di Milano: time-domain diffuse optical tomography.

Open Source & Tools

Open source · remote-gpu-trainer, an agent skill for Claude Code and Codex that runs, verifies and delivers GPU experiments (60+ GitHub stars)
Tools & lab · PyTorch, MATLAB, COMSOL; DMD/SLM illumination, single-pixel detection, optical alignment and system integration
Device modeling · Electro-optic phased-array geometry modeling and optimization for beam steering

Award & Service

2024 · Second Prize, National Post-Graduate Mathematical Contest in Modeling · Code
Reviewer · IEEE TIP, IEEE TCI, Optics Express, Optics Letters (13 verified reviews, 2025–2026)